

Subtraction of rational numbers

Add the opposite, exactly as with integers — the fractions change nothing.

LESSON 41–42

2 × 40 MIN

MATEMATIKA 7, §17

S8 7.2

LESSON CLOCK

15'

25'

25'

20'

5'

The rule	15 min
Different denominators	25 min
Mixed numbers	25 min
Differences on the line	20 min
Homework	5 min

LEARNING OBJECTIVES

- Subtract fractions with the same and with different denominators.
- Use $a - b = a + (-b)$ for rational numbers.
- Subtract mixed numbers, including with borrowing.
- Find the difference between two rational numbers on the line.

TEXTBOOK

National	pp. 108–113
Cambridge	Stage 8, pp. 70–76
Workbook	Exercise 7.2

01

Explanation

The rule

$$a - b = a + (-b)$$

Exactly as for integers. With like denominators, subtract the numerators:

$$\frac{a}{c} - \frac{b}{c} = \frac{a - b}{c}$$

DIFFERENCE	WORKING	ANSWER
$\frac{5}{7} - \frac{2}{7}$	$\frac{3}{7}$	$\frac{3}{7}$
$\frac{2}{7} - \frac{5}{7}$	$\frac{-3}{7}$	$-\frac{3}{7}$
$\frac{1}{4} - (-\frac{1}{4})$	$\frac{1}{4} + \frac{1}{4}$	$\frac{1}{2}$

TWO MINUSES GIVE A PLUS HERE TOO

$\frac{1}{4} - (-\frac{1}{4})$ is $\frac{1}{2}$. The rules of Chapter II apply to every rational number, not only to integers.

Different denominators

Same four steps as for addition: LCM, rewrite, subtract, simplify.

Example. $\frac{5}{6} - \frac{3}{4}$. LCM is 12:

$$\frac{10}{12} - \frac{9}{12} = \frac{1}{12}$$

DIFFERENCE	COMMON DENOMINATOR	ANSWER
$\frac{3}{4} - \frac{1}{6}$	$\frac{9}{12} - \frac{2}{12}$	$\frac{7}{12}$
$\frac{1}{3} - \frac{1}{2}$	$\frac{2}{6} - \frac{3}{6}$	$-\frac{1}{6}$
$-\frac{2}{5} - \frac{3}{10}$	$-\frac{4}{10} - \frac{3}{10}$	$-\frac{7}{10}$

A NEGATIVE ANSWER IS NOT A MISTAKE

$\frac{1}{3} - \frac{1}{2} = -\frac{1}{6}$ because a third is less than a half. Estimating which is bigger before subtracting tells you the sign in advance.

Mixed numbers

Two routes, as for addition. Route A — convert to improper fractions — never fails.

Example. $4\frac{1}{5} - 1\frac{3}{5}$.

ROUTE	WORKING
A	$\frac{21}{5} - \frac{8}{5} = \frac{13}{5} = 2\frac{3}{5}$
B	$4\frac{1}{5} = 3\frac{6}{5}$

Route B needs **borrowing**: since $\frac{1}{5}$ is smaller than $\frac{3}{5}$, one whole is exchanged for $\frac{5}{5}$. Both routes give $2\frac{3}{5}$.

BORROWING IS WHERE THE ERRORS ARE

$4\frac{1}{5}$ becomes $3\frac{6}{5}$, not $3\frac{11}{5}$. Converting to improper fractions avoids the step entirely, and is worth the extra line.

Differences on the line

The distance between two rational numbers is $|a - b|$, exactly as for integers.

POINTS	DIFFERENCE	DISTANCE
$\frac{3}{4}, \frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{2}$
$-\frac{1}{2}, \frac{1}{3}$	$-\frac{5}{6}$	$\frac{5}{6}$
$-\frac{3}{4}, -\frac{1}{4}$	$-\frac{1}{2}$	$\frac{1}{2}$

WHICH IS LARGER? SUBTRACT AND LOOK AT THE SIGN

A positive difference means the first is larger; a negative one means the second is. That is often quicker than finding a common denominator twice.

02 Worked examples

EXAMPLE 1 Compute $\frac{5}{6} - \frac{3}{4}$.

① LCM of 6 and 4 is 12.

② $\frac{10}{12} - \frac{9}{12}$

③ $= \frac{1}{12}$

④ Positive, so $\frac{5}{6}$ is the larger.

ANSWER

$$\frac{1}{12}$$

EXAMPLE 2 Compute $4\frac{1}{5} - 1\frac{3}{5}$.

① Convert: $\frac{21}{5} - \frac{8}{5}$.

Route A.

② $= \frac{13}{5}$

③ $= 2\frac{3}{5}$

④ Check: $2\frac{3}{5} + 1\frac{3}{5} = 4\frac{1}{5} \checkmark$

ANSWER

$$2\frac{3}{5}$$

EXAMPLE 3 Find the distance between $-\frac{1}{2}$ and $\frac{1}{3}$ on the line.

① $-\frac{1}{2} - \frac{1}{3}$

② $= -\frac{3}{6} - \frac{2}{6} = -\frac{5}{6}$

③ Take the modulus.

④ $\frac{5}{6}$

ANSWER

$$\frac{5}{6}$$

03

Interactive model

Ask which is larger, two thirds or three quarters, before any calculation; the class guesses, then subtracts, and the sign settles it.

Subtracting fractions

DENOMINATOR	FACTORISED	EXTRA FACTOR NEEDED
10	$2 \cdot 5$	2
4	2^2	5

LOWEST COMMON DENOMINATOR

20

$\frac{14}{20} - \frac{15}{20} = -\frac{1}{20}$ — the answer is negative.

04

Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Subtraction	Ayirish	Вычитание
Difference	Ayirma	Разность
Common denominator	Umumiy maxraj	Общий знаменатель
To borrow	Qarz olish	Занять
Improper fraction	Noto'g'ri kasr	Неправильная дробь
Opposite	Qarama-qarshi	Противоположное
Simplify	Soddalashtirish	Упростить
Compare	Taqqoslash	Сравнить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$\frac{2}{7} - \frac{5}{7}$ equals:

$$\frac{3}{7}$$

$$-\frac{3}{7}$$

$$\frac{7}{7}$$

$$-\frac{7}{7}$$

$\frac{1}{4} - (-\frac{1}{4})$ equals:

$$0$$

$$\frac{1}{2}$$

$$-\frac{1}{2}$$

$$\frac{1}{4}$$

$\underline{1} \frac{3}{6} - \underline{1} \frac{2}{6}$ equals:

$$\frac{1}{6}$$

$$-\frac{1}{6}$$

$$\frac{1}{5}$$

$$-\frac{1}{5}$$

$4 \underline{1} \frac{5}{5}$ with one whole borrowed:

$$3 \frac{6}{5}$$

$$3 \frac{11}{5}$$

$$3 \frac{1}{5}$$

$$5 \frac{1}{5}$$

The distance between $-\frac{1}{2}$ and $\frac{1}{3}$:

$$\frac{1}{6}$$

$$\frac{5}{6}$$

$$-\frac{5}{6}$$

$$\frac{2}{5}$$

A negative difference means:

an error

the second is larger

the first is larger

they are equal

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\frac{5}{7} - \frac{2}{7}$

ANSWER $\frac{3}{7}$

2. $\frac{2}{7} - \frac{5}{7}$

ANSWER $-\frac{3}{7}$

3. $\frac{3}{4} - \frac{1}{4}$

ANSWER $\frac{1}{2}$

4. $\frac{1}{4} - (-\frac{1}{4})$

ANSWER $\frac{1}{2}$

5. $\frac{1}{2} - \frac{1}{4}$

ANSWER $\frac{1}{4}$

6. $0.8 - 0.35$

ANSWER 0.45

7. $\frac{5}{8} - \frac{1}{8}$

ANSWER $\frac{1}{2}$

Medium · the standard question

7 PROBLEMS

1. $\frac{5}{6} - \frac{3}{4}$

ANSWER $\frac{1}{12}$

2. $\frac{3}{4} - \frac{1}{6}$

ANSWER $\frac{7}{12}$

3. $\frac{1}{3} - \frac{1}{2}$

ANSWER $-\frac{1}{6}$

4. $-\frac{2}{5} - \frac{3}{10}$

ANSWER $-\frac{7}{10}$

5. $4\frac{1}{5} - 1\frac{3}{5}$

ANSWER $2\frac{3}{5}$

6. $3\frac{1}{4} - 1\frac{1}{2}$

ANSWER $1\frac{3}{4}$

7. Distance between $-\frac{1}{2}$ and $\frac{1}{3}$

ANSWER $\frac{5}{6}$

Hard · stretch and reason

7 PROBLEMS

1. $\frac{7}{8} - \frac{2}{3} - \frac{1}{6}$

ANSWER $\frac{1}{24}$

2. $-1\frac{1}{3} - (-2\frac{1}{2})$

ANSWER $1\frac{1}{6}$

3. $5 - 2\frac{3}{7}$

ANSWER $2\frac{4}{7}$

4. Find x : $\frac{3}{4} - x = \frac{1}{3}$

ANSWER $\frac{5}{12}$

5. Which is larger, $\frac{5}{8}$ or $\frac{7}{11}$?

ANSWER $\frac{7}{11}$

6. The difference between the largest and smallest of $\frac{1}{2}$, $-\frac{2}{3}$, $\frac{3}{4}$

ANSWER $\frac{17}{12}$

7. $0.6 - \frac{1}{4}$

ANSWER $\frac{7}{20}$

07 Homework**Homework — 5 tasks**

Predict the sign of every answer before computing it.

1. Compute $\frac{7}{9} - \frac{2}{3}$ and $\frac{2}{5} - \frac{3}{4}$.
2. Compute $-\frac{1}{2} - \frac{2}{3}$.
3. Compute $5\frac{1}{4} - 2\frac{2}{3}$.
4. Find the distance between $-\frac{3}{4}$ and $\frac{1}{2}$.
5. Find x if $\frac{2}{3} - x = \frac{1}{6}$.